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Quantitative analytical model for magnetic reconnection in Hall magnetohydrodynamics

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Abstract

Magnetic reconnection is of fundamental importance for laboratory and naturally occurring plasmas. Reconnection usually develops on time scales which are much shorter than those associated with classical collisional dissipation processes, and which are not fully understood. While such dissipation-coefficient-independent (or "fast") reconnection rates have been observed in particle and Hall magnetohydrodynamics (MHD) simulations and predicted analytically in electron MHD, a quantitative analytical theory of fast reconnection valid for arbitrary ion inertial lengths, d_i, has been lacking. Here we propose such a theory without a guide field. The theory describes two-dimensional magnetic field diffusion regions, provides expressions for the reconnection rates, and derives a formal criterion for fast reconnection in terms of dissipation parameters and d_i. It also resolves existing controversies by demonstrating that both open X-point and elongated diffusion regions allow fast reconnection, and by revealing a possibility of strong dependence of the reconnection rates on d_i.

1 INTRODUCTION

Magnetic reconnection [1, 2] – the topological rearrangement of magnetic fields immersed in highly electrically conductive plasma and the accompanying magnetic energy release – is responsible for many natural phenomena, such as Earth magnetospheric substorms [1, 3]; solar flares, coronal mass ejections, and coronal heating [1, 4]; stellar flares [5]; protostellar collapses [6]; the generation of large-scale magnetic fields in radio lobes of active galactic nuclei [7]; and the generation of energetic particles, e.g. during solar flares [8] and magnetospheric substorms [9]. In laboratory plasmas, reconnection is inevitably present in nuclear fusion magnetic confinement configurations. Some of these configurations suffer from plasma instabilities caused by reconnection (e.g. sawtooth oscillations and some types of major disruptions in tokamaks [10]), while others rely heavily upon it to create and even sustain the confining magnetic field through a dynamo action (e.g. reversed field pinches and spheromaks [11]).

The extremely fast time scales observed for reconnection phenomena are consistent with reconnection rates being independent of collisional dissipation coefficients (e.g. resistivity). Such "fast" reconnection rates have been observed computationally in fluid [Hall magnetohydrodynamics (MHD)] and particle simulations [12, 13, 14, 15, 16, 17]. However, a comprehensive analytical theory capable of explaining them has been lacking. Hall MHD is arguably the simplest fluid description of a magnetized plasma that features fast reconnection [17]. An important physical parameter in Hall MHD reconnection is the ratio of the ion inertial length, d_i (defined later in this paper), to the magnetic field diffusion layer thickness, δ. Existing quantitative analytical models have only concentrated on limiting regimes of Hall MHD, such as resistive MHD ($d_i \ll \delta$) [18, 19, 20] and electron MHD [21] ($d_i \gg \delta$). None is valid

in general. Resistive MHD models predict reconnection rates that are either too slow (e.g. the Sweet-Parker model [18, 19]), or require overly restrictive assumptions for their validity [22] (e.g. the Petschek model [20]). A novel electron MHD model [21], developed recently by the present authors, does predict fast reconnection, but assumes that ions have no dynamics. A quantitative analytical model valid in all regimes of Hall MHD (arbitrary d_i/δ) is essential to explain observations, and to describe transitions between these different regimes.

This work presents, for the first time, a simple analytical theory of magnetic reconnection in all d_i regimes. The theory describes the magnetic field diffusion region in two dimensions. It considers steady-state reconnection *without a guide field*, and takes into account both plasma resistivity and electron viscosity (sometimes referred to as hyper-resistivity). Although a guide field is often finite in physical systems, understanding a zero-guide-field limit has been an important milestone both theoretically (e.g. the Geospace Environmental Modeling (GEM) Magnetic Reconnection Challenge [17]) and experimentally (e.g. the leading US reconnection experiment MRX [23]). Moreover, the theory discussed herein can be straightforwardly extended to include finite guide field effects [24], which is a subject of future work.

Our theory recovers resistive MHD (Sweet-Parker [18, 19]) and electron MHD results [21] in the appropriate limits, and is valid everywhere in-between. It yields the diffusion region aspect ratio, plasma flows, magnetic field components, and the reconnection rate as functions of inertial and dissipation parameters. In particular, for given dissipation, it predicts the d_i-threshold for which reconnection rates become dissipation-independent. We have validated the theoretical predictions with thirty five non-linear simulations of the magnetic island coalescence problem [25, 26, 27, 28, 29, 30, 31, 32, 33, 34], with resistivity, hyper-resistivity, and ion inertial length

varying over many orders of magnitude each.

Our theory confirms a number of long-standing empirical results, such as the possibility of fast reconnection for moderate d_i ($d_i \gtrsim \delta$) in Hall MHD [12, 13, 14, 15, 16, 17], and the independence of the reconnection rate of the dissipation mechanism [15]. It also resolves some outstanding controversies in Hall MHD reconnection. In particular, it shows that both open X-point [17] and elongated [36] diffusion regions can result in fast reconnection, and that, contrary to previous claims [37], the reconnection rate can exhibit a strong dependence on the ion inertial length d_i [38, 39].

A brief summary of our work was given in a recent Physical Review Letter [35], where a number of significant details and important results were omitted due to space limitations. Herein, we present a more complete account of our new theory, including equation derivations, tables of numerical results, a number of additional figures, and an intuitive physical picture for fast reconnection.

This paper is organized as follows. Section 2 briefly summarizes Hall MHD model and introduces key plasma quantities (discrete variables) necessary for describing microscopic reconnection region. Section 3 derives dynamical equations for these quantities. It is realized in Sec. 4 that steady-state analysis is normally appropriate for describing reconnection at and around the moment of the local maximum of the reconnection rate, and that the aspect ratio ξ of the electron diffusion region is a key quantity for understanding reconnection. Then, steady-state limits of the dynamical equations from Sec. 3 are used to obtain an equation for ξ valid for arbitrary d_i. Resistive and electron MHD limits of the solution for ξ and the corresponding reconnection rate expressions are obtained in Secs. 5 and 6, respectively, while the general solution is obtained in Sec. 7. The analytical predictions are thoroughly benchmarked with two-dimensional fully-nonlinear numerical simulations of the island coalescence

instability in Sec. 8 and a simple intuitive physical picture of fast reconnection is presented in Sec. 9. Finally, the results are summarized and the conclusions are drawn in Sec. 10.

2 INCOMPRESSIBLE HALL MHD MODEL

Hall MHD is arguably the simplest fluid description of a magnetized plasma that captures differences in behavior between electrons and ions, and is thereby capable of producing fast reconnection. Like resistive MHD, Hall MHD describes plasma as a single magnetized fluid on length scales large compared with d_i (i.e., outside the *ion inertial region*). And like electron MHD, Hall MHD describes plasma as a magnetized electron fluid in the presence of a passive, demagnetized, neutralizing ion population on length scales small compared with d_i (i.e., inside the ion inertial region). The electron fluid also eventually becomes demagnetized inside the *electron diffusion region*, which lies within the ion inertial region and can be much smaller. The shape and size of the electron diffusion region plays a crucial role in determining the reconnection rate. Electron inertia can also be an important mechanism for demagnetization of the electron fluid, but is not considered in this work.

Assuming that plasma pressure exceeds magnetic pressure, so that plasma density, n, is constant in space and time, and normalizing to the Alfvén speed, v_A, and an arbitrary length (system size), L, the incompressible Hall MHD equations become [2]

$$\partial_t \boldsymbol{B} + \boldsymbol{\nabla} \times (\boldsymbol{B} \times \boldsymbol{V}_e) = -\eta \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{B}) + \eta_H \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \nabla^2 \boldsymbol{B}), \tag{1}$$

$$\partial_t \boldsymbol{V} + \boldsymbol{V} \cdot \boldsymbol{\nabla} \boldsymbol{V} + \boldsymbol{\nabla} p = \boldsymbol{B} \cdot \boldsymbol{\nabla} \boldsymbol{B}, \quad \boldsymbol{\nabla} \cdot \boldsymbol{V} = 0. \tag{2}$$

Here, $\boldsymbol{B}$ is the magnetic field, $\boldsymbol{V}_e \equiv \boldsymbol{V} - d_i \boldsymbol{J}$ is the electron flow velocity (with $\boldsymbol{V}$ the ion flow velocity and $\boldsymbol{J} = \boldsymbol{\nabla} \times \boldsymbol{B}$ the total current), and p is the sum of magnetic and

plasma kinetic pressures, which is eliminated by employing the $\boldsymbol{\nabla} \cdot \boldsymbol{V} = 0$ constraint. The ion inertial length is $d_i \equiv c/(\omega_{pi} L)$, with c the speed of light, $\omega_{pi} \equiv (4\pi n e^2/M)^{1/2}$ the ion plasma frequency, e the unit electric charge, and M the ion mass. Finally, $\eta \ll 1$ and $\eta_H \ll 1$ are the normalized resistivity and electron viscosity (hyper-resistivity), respectively.

This study concentrates on the reconnection region and does not consider a particular macroscopic physical system supplying magnetic flux. A fully closed dynamical description can be obtained by coupling our model equations with a suitable macroscopic driver, as done in Ref. [34]. However, this is out of the scope of this work.

In what follows, we assume the two-dimensional reconnection geometry shown in Fig. 1, with the ignorable direction along the z axis, so that $\partial/\partial z = 0$. The electron diffusion region is approximated with the smaller green rectangle of (normalized) dimensions δ and w, and the ion inertial region with the larger grey rectangle of dimensions $h \geq \delta$ and $\Delta \geq w$. We assume that Δ corresponds to an external length scale (e.g. the magnetic island width in the island coalescence problem [25, 26, 27, 28, 29, 30, 31, 32, 33, 34]) and is not a dynamical quantity. To be able to recover the limiting case of resistive MHD (when the two regions coincide), we assume

$$h \equiv \max(d_i, \delta), \tag{3}$$

since no electron-ion decoupling is expected to occur for $d_i < \delta$ [25, 40, 41]. For the same reason, we take $w = \Delta$ whenever $h = \delta$.

To describe the reconnection region, it is convenient to use discrete electron and ion flow and magnetic field variables, as shown in Fig. 1. Specifically, we introduce "outside" flow variables $V_y^{out} \equiv V_y(0, h/2)$, $V_{e,y}^{out} \equiv V_{e,y}(0, h/2)$, $V_x^{out} \equiv V_x(\Delta/2, 0)$, and $V_{e,x}^{out} \equiv V_{e,x}(\Delta/2, 0)$; "inside" flow variables $V_y^{in} \equiv V_y(0, \delta/2)$, $V_{e,y}^{in} \equiv V_{e,y}(0, \delta/2)$, $V_x^{in} \equiv V_x(w/2, 0)$, and $V_{e,x}^{in} \equiv V_{e,x}(w/2, 0)$; and magnetic field variables $B_x^{out} \equiv$

$B_x(0,h/2)$, $B_y^{out} \equiv B_y(\Delta/2,0)$, $B_x^{in} \equiv B_x(0,\delta/2)$, $B_y^{in} \equiv B_y(w/2,0)$, and $B_z^{in} \equiv -B_z(w/2,\delta/2)$.

Electrons and ions enter and exit the ion inertial region with approximately equal velocities: $V_y^{out} \approx V_{e,y}^{out}$ and $V_x^{out} \approx V_{e,x}^{out}$. The species decouple within the ion inertial region, creating in-plane currents and a quadrupolar magnetic field B_z^{in}. The ion flow is negligible within the electron diffusion region: $V_y^{in} \ll V_{e,y}^{in}$ and $V_x^{in} \ll V_{e,x}^{in}$.

3 EQUATIONS FOR THE DISCRETE VARIABLES

Dynamical equations for the discrete flow and magnetic field variables introduced in the previous section can be obtained from Hall MHD Eqs. (1) and (2). Notice, that the z components of Eqs. (1) and (2) predict that B_z is coupled with the ion flow in the z direction. However, based on the numerical results for Hall MHD reconnection [25], V_z is expected to be small, $V_z \ll V_{e,z}$, and will be neglected.

3.1 Inner magnetic field variables

First, we follow Refs. [21, 34] to derive equations for the inner magnetic field variables B_x^{in}, B_y^{in}, and B_z^{in}. An equation for B_x^{in} is conveniently obtained from the x component of Eq. (1),

$$\partial_t B_x + \partial_y(V_{e,y}B_x - B_y V_{e,x}) = -(\eta - \eta_H \nabla^2)(\partial_{xy}^2 B_y - \partial_y^2 B_x), \tag{4}$$

evaluated at $(x,y) = (0,\delta/2)$. Using $\partial_y \sim 1/\delta$ gives [21, 34] $\partial_t B_x|_{(0,\delta/2)} \sim \mathrm{d}_t B_x^{in} - B_x^{in}\mathrm{d}_t \ln\delta$, where d_t denotes the total time derivative, the second term accounts for the time variation of δ, and we neglect numerical factors of order unity. Similarly, $\partial_y(V_{e,y}B_x - V_{e,x}B_y)|_{(0,\delta/2)} \sim -B_x^{in}V_{e,y}^{in}/\delta$ and $\partial_y^2 B_x|_{(0,\delta/2)} \sim -B_x^{in}/\delta^2$. Although $B_y|_{(0,\delta/2)} = 0$, $\partial_{xy}^2 B_y|_{(0,\delta/2)}$ is finite. To estimate this quantity, we notice

that $\partial_x B_y|_{(0,0)} \sim B_y^{in}/w$, since $\partial_x \sim 1/w$. Then, $\partial^2_{xy} B_y|_{(0,\delta/2)} \sim (\partial_x B_y|_{(0,\delta/2)} - \partial_x B_y|_{(0,0)})/\delta \sim -\partial_x B_y|_{(0,0)}/\delta \sim -B_y^{in}/(\delta w)$. Finally, $\nabla^2 \sim -(\delta^{-2} + w^{-2})$. Putting everything together gives the following equation for B_x^{in}:

$$\mathrm{d}_t B_x^{in} - \frac{B_x^{in}(\mathrm{d}_t \delta + V_{e,y}^{in})}{\delta} = \eta_{\mathrm{eff}} \left(\frac{B_y^{in}}{\delta w} - \frac{B_x^{in}}{\delta^2} \right), \tag{5}$$

where $\eta_{\mathrm{eff}} \equiv \eta + \mu \eta_H (\delta^{-2} + w^{-2})$. In this and the other discrete equations, we neglect for simplicity all numerical factors of order unity. The only numerical factor retained, μ, describes relative importance of resistive and hyper-resistive terms and is of order $\nabla^2 J_z / J_z$, as evaluated at the half width of J_z. If we assume $J_z \propto \exp[-(2x/\delta)^2 - (2y/w)^2]$ then $\mu = 8$, whereas if we take $J_z \propto \exp[-(2x/\delta)^4 - (2y/w)^4]$ then $\mu = 16$. Therefore, it appears that the correct value for μ is on the order of 10 or 20. In this work, we employ $\mu = 16$.

An equation for B_y^{in} can be obtained in a similar manner from the y component of Eq. (1) evaluated at $(x, y) = (w/2, 0)$. However, there is an easier way. Since reversing the electron flow simply interchanges roles of B_x^{in} and B_y^{in}, and the direction of the in-plane current is determined by the sigh of B_z, equations for B_x^{in}, B_y^{in}, and B_z^{in} must be invariant under the substitution

$$(B_x^{in}, B_y^{in}, B_z^{in}, V_{e,x}^{in}, V_{e,y}^{in}, \delta, w) \leftrightarrow (B_y^{in}, B_x^{in}, -B_z^{in}, -V_{e,y}^{in}, -V_{e,x}^{in}, w, \delta). \tag{6}$$

Using substitution (6) in Eq. (5) gives the following equation for B_y^{in}:

$$\mathrm{d}_t B_y^{in} - \frac{B_y^{in}(\mathrm{d}_t w - V_{e,x}^{in})}{w} = \eta_{\mathrm{eff}} \left(\frac{B_x^{in}}{\delta w} - \frac{B_y^{in}}{w^2} \right). \tag{7}$$

An equation for B_z^{in} is obtained from the z component of Eq. (1),

$$\partial_t B_z + (V_x \partial_x + V_y \partial_y) B_z + d_i (B_x \partial_x + B_y \partial_y) J_z = -(\eta - \eta_H \nabla^2) \nabla^2 B_z, \tag{8}$$

evaluated at $(x, y) = (w/2, \delta/2)$. Similarly to the B_x^{in} case, $\partial_t B_z|_{(w/2,\delta/2)} \sim -\mathrm{d}_t B_z^{in} + B_z^{in}(\mathrm{d}_t \ln \delta + \mathrm{d}_t \ln w)$. Employing $\partial_x \sim 1/w$, $\partial_y \sim 1/\delta$, and plasma flow incompress-

ibility (2) gives $(V_x\partial_x + V_y\partial_y)B_z|_{(w/2,\delta/2)} \sim B_z^{in}(V_y^{in}/\delta - V_x^{in}/w) = 0$. Noticing that $J_z = \partial_x B_y - \partial_y B_x$ results in $(B_x\partial_x + B_y\partial_y)J_z|_{(w/2,\delta/2)} \sim -(B_x^{in}/w + B_y^{in}/\delta)J_z|_{(w/2,\delta/2)} \sim -(B_x^{in}/w + B_y^{in}/\delta)(B_y^{in}/w - B_x^{in}/\delta)$. Finally, employing $\nabla^2 \sim -(\delta^{-2} + w^{-2})$ and putting everything together gives the following equation for B_z^{in}:

$$\begin{aligned} &\mathrm{d}_t B_z^{in} + d_i\left(\frac{B_x^{in}}{w} + \frac{B_y^{in}}{\delta}\right)\left(\frac{B_y^{in}}{w} - \frac{B_x^{in}}{\delta}\right) \\ &-B_z^{in}\left(\frac{\mathrm{d}_t\delta}{\delta} + \frac{\mathrm{d}_t w}{w}\right) = -B_z^{in}\eta_{\mathrm{eff}}\left(\frac{1}{\delta^2} + \frac{1}{w^2}\right), \end{aligned} \tag{9}$$

which is clearly invariant under substitution (6).

Collisional terms (i.e., right-hand sides) of Eqs. (5), (7), and (9) contain both sinks and sources. The sinks correspond to dissipation of magnetic field components, while the sources describe the fact that reconnection of a certain component of the magnetic field results in creation of the orthogonal component (e.g. reconnection of B_x^{in} creates B_y^{in}). Capturing these effects in discrete equations is most conveniently done by retaining $\boldsymbol{\nabla}\times\boldsymbol{\nabla}\times\boldsymbol{B}$ terms instead of $-\nabla^2\boldsymbol{B}$ in the continuum equation (1). Note that our equations for B_x^{in}, B_y^{in}, and B_z^{in} correctly dissipate magnetic energy [21].

3.2 Outer magnetic field variables

Equations for B_x^{out} and B_y^{out} are obtained in a similar fashion by discretizing the x and y components of Eq. (1) with dissipation neglected,

$$\begin{aligned} \partial_t B_x + \partial_y(V_{e,y}B_x - B_y V_{e,x}) &= 0, \\ \partial_t B_y + \partial_x(V_{e,x}B_y - B_x V_{e,y}) &= 0, \end{aligned} \tag{10}$$

at $(x,y) = (0, h/2)$ and $(\Delta/2, 0)$, respectively. Noticing that $\partial_t B_x|_{(0,h/2)} \sim \mathrm{d}_t B_x^{out} - \mathrm{d}_t h(B_x^{out} - B_x^{in})/(h-\delta)$ and $\partial_y(V_{e,y}B_x - B_y V_{e,x})|_{(0,h/2)} \sim (V_{e,y}^{out}B_x^{out} - V_{e,y}^{in}B_x^{in})/(h-\delta)$

gives the following equation for B_x^{out}:

$$\mathrm{d}_t B_x^{out} = \mathrm{d}_t h \frac{B_x^{out} - B_x^{in}}{h - \delta} + \frac{V_{e,y}^{out} B_x^{out} - V_{e,y}^{in} B_x^{in}}{h - \delta}. \tag{11}$$

Similarly,

$$\mathrm{d}_t B_y^{out} = -\frac{V_{e,x}^{out} B_y^{out} - V_{e,x}^{in} B_y^{in}}{\Delta - w}. \tag{12}$$

A $\mathrm{d}_t \Delta$ term is absent from equation for B_y^{out} since Δ is assumed time independent.

3.3 Electron flow variables

Next, we employ Eq. (2) and the relation between $\boldsymbol{V}$ and $\boldsymbol{V}_e$ to obtain time evolution equations for electron flow variables. Using $\boldsymbol{V}_e = \boldsymbol{V} - d_i \boldsymbol{\nabla} \times \boldsymbol{B}$, and assuming that the ion flow velocity decreases linearly from the boundary towards the center of the ion inertial region, gives the following equations for $V_{e,x}^{in}$ and $V_{e,y}^{in}$:

$$V_{e,x}^{in} = V_x^{in} + \frac{d_i B_z^{in}}{\delta} = V_x^{out} \frac{w}{\Delta} + \frac{d_i B_z^{in}}{\delta}, \tag{13}$$

$$V_{e,y}^{in} = V_y^{in} + \frac{d_i B_z^{in}}{w} = V_y^{out} \frac{\delta}{h} + \frac{d_i B_z^{in}}{w}. \tag{14}$$

According to numerical simulations [25] and Fig. 2c, B_z is approximately constant along magnetic separatrix everywhere inside the ion inertial region (but outside the electron diffusion region). Therefore, we take $B_z(w/2, \delta/2) \approx B_z(\Delta/2, h/2) = -B_z^{in}$. Physically, this follows from the fact that electrons are magnetized everywhere outside the electron diffusion region and cannot cross the magnetic separatrix, while ions become demagnetized inside the ion inertial region and do cross the separatrix there. This creates in-plane currents flowing in narrow layers along the separatrix inside the ion inertial (but outside the electron diffusion) region. Taking this fact into account,

we can relate electron and ion flows at the ion inertial region boundaries:

$$V_{e,x}^{out} = V_{i,x}^{out} + \frac{d_i B_z^{in}}{h}, \tag{15}$$

$$V_{e,y}^{out} = V_{i,y}^{out} + \frac{d_i B_z^{in}}{\Delta}. \tag{16}$$

3.4 Ion flow variables

To close the system of discrete equations we also require time evolution equations for the ion flow components V_x^{out} and V_y^{out}, which are obtained from the x component of plasma momentum equation, Eq. (2), and incompressibility condition $\boldsymbol{\nabla} \cdot \boldsymbol{V} = 0$, respectively. The derivation is described in detail in Ref. [34]. The resulting equations are

$$\mathrm{d}_t V_x^{out} - \frac{\mathrm{d}_t w}{w} V_x^{out} = \frac{\Delta^2}{h(h^2+\Delta^2)} \Bigg[(B_y^{out})^2 - \frac{h^2}{\Delta^2} (B_x^{out})^2 + \frac{h}{\Delta} \frac{(V_y^{out})^2 + (B_x^{out})^2 - (V_x^{out})^2 - (B_y^{out})^2}{2} \Bigg], \tag{17}$$

$$V_y^{out} = V_x^{out} \frac{h}{\Delta}. \tag{18}$$

Together with Eq. (3), the discrete Eqs. (5), (7), (9), (11), (12) – (18) comprise a system of twelve equations for fourteen unknowns, h, δ, w, B_x^{in}, B_y^{in}, B_x^{out}, B_y^{out}, B_z^{in}, $V_{e,x}^{in}$, $V_{e,y}^{in}$, $V_{e,x}^{out}$, $V_{e,y}^{out}$, V_x^{out}, and V_y^{out}.

4 STEADY-STATE ANALYSIS

Time-derivative terms in Eqs. (1) and (2) (and thereby in the discrete equations) are normally found small at and around the time of the local maximum of the reconnection rate [34], and a steady-state analysis is appropriate in such situations. We proceed to perform such a steady-state analysis for the discrete electron and ion flow and magnetic field equations by setting $\mathrm{d}_t = 0$.

An important characteristic of the reconnection region is the aspect ratio of the electron diffusion region, $\xi \equiv \delta/w$. The reconnection rate, $E_z = -(\eta - \eta_H \nabla^2) J_z = [\eta + \mu\eta_H(\delta^{-2} + w^{-2})](B_x^{in}/\delta - B_y^{in}/w)$, can be conveniently normalized to $(B_x^{in})^2$ and written in terms of ξ to become

$$E_{z*} \equiv \frac{E_z}{(B_x^{in})^2} = 2f(\xi)^2 \left(\frac{1-\xi^2}{\xi}\right), \tag{19}$$

where $f(\xi) \equiv [\sqrt{2} w S(\xi)]^{-1/2}$. The effective Lundquist number $S(\xi)$ is defined as $S^{-1}(\xi) \equiv S_\eta^{-1} + S_H^{-1}(1+\xi^{-2})$, where $S_\eta \equiv \sqrt{2} B_x^{in}/\eta$ and $S_H \equiv \sqrt{2} B_x^{in} w^2/(\mu\eta_H)$ are the resistive and the hyper-resistive Lundquist numbers, respectively. We assume that collisional dissipation is small, so that corresponding Lundquist numbers are large: $S_\eta \gg 1$, $S_H \gg 1$, and $S(\xi) \gg 1$. It is clear from Eq. (19) that electron diffusion regions with $\xi^2 \ll 1$ are expected to result in larger E_{z*} and faster magnetic energy release than those with $\xi \sim 1$. Therefore, we will concentrate on the limit $\xi^2 \ll 1$ (but allow $\xi < 1$) in the remainder of this work, by approximating $1-\xi^2 \approx 1+\xi^2 \approx 1$.

Next, we obtain an equation for ξ. It is convenient to write such an equation in terms of normalized quantities $\xi_* \equiv \xi/f(\xi)$ and $d_* \equiv d_i/[f(\xi)w]$. Employing the steady-state discrete equations, Eqs. (3), (5), (7), (9), (11), (12) – (18), to eliminate variables in favor of ξ, gives:

$$\alpha \left(1 - \frac{d_*^2 \xi_*^2}{2}\right)^2 + 1 - \frac{(1+d_*^2)\xi_*^2}{2} = 0. \tag{20}$$

Here, $\alpha \equiv h\Delta/(\delta w) - 1$ is a measure of two-fluid separation, with $h\Delta/(\delta w)$ the ratio of areas of the ion inertial over the electron diffusion regions. Equation (20) contains all relevant information about the diffusion region geometry, in all regimes of interest, and is one of the main results of this study.

Other plasma quantities of interest can be conveniently expressed in terms of α, ξ, ξ_*, and d_*. In particular, generated magnetic field components B_y^{in} and B_z^{in} are

given by

$$\frac{B_y^{in}}{B_x^{in}} = \xi, \quad \frac{B_z^{in}}{B_x^{in}} = \frac{1 + 2\alpha - \sqrt{1 + 2\alpha\xi_*^2}}{\alpha d_* \xi_*}, \tag{21}$$

and the outgoing outer ion and inner electron flow velocities read:

$$\frac{V_x^{out}}{B_x^{in}} = \frac{\sqrt{1 + 2\alpha\xi_*^2} - 1}{\alpha\xi_*^2}, \quad \frac{V_{e,x}^{in}}{B_x^{in}} = \frac{2}{\xi_*^2}. \tag{22}$$

The magnitude of α (but not its exact value) clearly determines the asymptotic solution of Eq. (20). We expect to have $\alpha \ll 1$ (and the first term to be negligible) in the resistive MHD limit and $\alpha \gg 1$ (and the last two terms to be negligible) in the electron MHD limit. It is convenient to consider these two limits before discussing the general solution of Eq. (20).

5 RESISTIVE MHD LIMIT

We begin by considering the situation when electron diffusion and ion inertial regions essentially coincide ($\alpha \ll 1$, and $w \approx \Delta$), and electron and ion motions do not separate. Taking this limit in Eq. (20) gives the diffusion region aspect ratio

$$\xi_* \approx \sqrt{\frac{2}{1 + d_*^2}} \quad \Rightarrow \quad \xi \approx \sqrt{\frac{\sqrt{2}}{\Delta S(\xi) + \sqrt{2} d_i^2 S(\xi)^2}}. \tag{23}$$

The standard resistive MHD result [18, 19, 34] is obtained by letting $d_i \to 0$: $\xi \approx [\Delta S(\xi)/\sqrt{2}]^{-1/2}$. For negligible hyper-resistivity, this gives $\xi \approx [\Delta S_\eta/\sqrt{2}]^{-1/2} \ll 1$ and the reconnection region is always fairly elongated. At the same time, for negligible resistivity, we obtain $\xi \approx [\Delta S_H/\sqrt{2}]^{-1/4}$. Note that the scaling of ξ with S_H is weaker than with S_η, and less-elongated reconnection regions are allowed even for $S_H \gg S_\eta$. In both cases, the reconnection rate (19) $E_{z*} \approx \xi \ll 1$ is slow and exhibits strong dependence on dissipation coefficients. Equation (21) predicts that

$B_z^{in} \to 0$ as $d_i \to 0$. Finally, according to Eq. (22), the outgoing flow is Alfvénic, $V_x^{out}/B_x^{out} \approx 1$, while the incoming flow velocity is equal to the reconnection rate, $V_y^{out}/B_x^{out} = \xi V_x^{out}/B_x^{out} \approx \xi = E_{z*}$.

6 ELECTRON MHD LIMIT

Next, we consider the situation when electron diffusion region becomes much smaller than the ion inertial region ($\alpha \gg 1$), and electron and ion in-plane motions separate. In this limit, we find from Eq. (20)

$$\xi_* \approx \frac{\sqrt{2}}{d_*} \quad \Rightarrow \quad \xi \approx \frac{1}{d_i S(\xi)}, \tag{24}$$

which recovers the electron MHD result obtained in Ref. [21]. In particular, for negligible hyper-resistivity, Eq. (24) gives $\xi \approx (d_i S_\eta)^{-1}$ and the electron diffusion region is more elongated than in resistive MHD. At the same time, for negligible resistivity, Eq. (24) predicts $\xi \approx (d_i S_H)^{-1/3}$. Since $S_H \propto w^2$, depending on the macroscopic driver this permits electron diffusion regions which are either elongated, $\delta \ll w \sim \Delta$, or have near-unity aspect ratios, $\delta \sim w \ll \Delta$ [21]. The reconnection rate evaluated from Eq. (19) is not explicitly dependent on dissipation coefficients, $E_{z*} \approx \sqrt{2} d_i/w$, and is therefore fast (recall $w \leq \Delta$). In particular, the reconnection rate is formally independent of the dissipation mechanism, and features a strong dependence on d_i. Moreover, E_{z*} does not depend on ξ, i.e., both square and elongated electron diffusion regions allow fast reconnection.

Using result (24) in Eq. (21) gives $B_z^{in}/B_x^{in} \approx \sqrt{2}$, in agreement with Ref. [21]. In addition, Eq. (22) predicts that the outgoing electron flow velocity increases with d_i, $V_{e,x}^{in}/B_x^{in} \approx d_*^2$, while the incoming electron flow velocity is equal to the reconnection rate, $V_{e,y}^{in}/B_x^{in} = \xi V_{e,x}^{in}/B_x^{in} = E_{z*}$. To avoid unphysically large outgoing electron flow

velocities $V_{e,x}^{in}$ at large d_i, electron inertia must be taken into account. This can be done by incorporating inertial effects into equations for B_x^{in}, B_y^{in}, and B_z^{in} and will be discussed in future work.

7 HALL MHD SOLUTION

Expressions (23) and (24) for ξ_* are valid for $\alpha \ll 1$ and $\alpha \gg 1$, respectively. However, we notice that they are close at $d_* = 1$, and that Eq. (23) smoothly transitions into Eq. (24) for $d_* \gg 1$. It follows that the range of d_* where finite-α effects are important is rather narrow. Moreover, none of the physical quantities of interest seem to vary much over this range. Consequently, it appears reasonable to approximate ξ for all d_i with Eq. (23) and the reconnection rate with

$$E_{z*} \approx \sqrt{2f(\xi)^2(1+d_*^2)} \approx \sqrt{\frac{\sqrt{2}}{wS(\xi)} + \frac{2d_i^2}{w^2}}. \tag{25}$$

These approximations do not involve α and reduce to correct resistive and electron MHD limits. Moreover, we shall show in the next section that Eqs. (23) and (25) are in excellent agrement with nonlinear numerical simulations in all regimes of interest.

Equation (25) predicts that "slow" (Sweet-Parker) reconnection occurs for $d_* \lesssim 1$, while fast (dissipation-independent) reconnection occurs for $d_* \gtrsim 3$. The natural threshold between slow and fast reconnection is $d_* = 1$. Using Eq. (23), this threshold can be written as $d_i = d_{ic} \equiv w\xi_c$, with ξ_c satisfying $\xi_c = f(\xi_c) = [\sqrt{2}wS(\xi_c)]^{-1/2}$. Then, reconnection is slow for $d_i \lesssim d_{ic}$ and fast for $d_i \gtrsim 3d_{ic}$. In practice, one can estimate d_{ic} from dissipation coefficients and w (for which the appropriate equilibrium length scale Δ can be used) and therefore determine *a priori* whether a given d_i will produce slow or fast reconnection. This threshold estimate is consistent with the intuitive criterion [25, 40, 41] that Hall effects are relevant when $\delta < d_i$, which is an

equivalent form of $d_* > 1$.

The out-of-plane magnetic field, which again can be written in the α-independent form by extrapolating resistive and electron MHD expressions to $d_* = 1$, $B_z^{in}/B_x^{in} = \sqrt{2d_*^2/(1+d_*^2)}$, is only negligible for $d_* \ll 1$. Consequently, even though B_z^{in} is finite at $d_* \lesssim 1$, its effect on the reconnection rate is not important [40] in this regime, so that finite B_z^{in} is not necessarily a signature of fast reconnection.

It is instructive to relate our findings (23) with recent studies [41, 42] of Hall MHD reconnection, which found that three distinct solutions for δ with distinct reconnection rates were possible in the vicinity of $d_* = 1$. These findings do not contradict our prediction that only one solution for ξ is possible. Indeed, Ref. [42] cites $\delta_1 = 0.4d_i$, $\delta_2 = 1.02d_i$, $\delta_3 = 2.44d_i$, and $w_2 = 60d_i$. We estimate from the figures in the reference that $w_1 \approx 28d_i$ and $w_3 \approx 152d_i$. As a result, $\xi_1 \approx \xi_2 \approx \xi_3 \approx 0.016$, which confirms that a single solution in ξ exists. It follows that existence of the bifurcated solutions in δ is related to the interaction between a microscopic diffusion region and a specific macroscopic driver. Understanding this interaction is out of the scope of this work.

8 NUMERICAL VALIDATION OF THE THEORY

We proceed to validate predictions of Eqs. (23) and (25) by comparing them with full two-dimensional numerical simulations [43] of the magnetic island coalescence instability [25, 26, 27, 28, 29, 30, 31, 32, 33, 34]. We slightly perturb the equilibrium given by the magnetic flux function

$$\psi(x,y) = -(2\pi)^{-1}\ln[\cosh(2\pi x) + \epsilon\cos(2\pi y)],$$

with $\epsilon = 0.2$, $-1 \leq x \leq 1$, and $-\infty < y < +\infty$, and study the evolution of the perturbation by numerically solving Eqs. (1) and (2). As usual, the magnetic field is given in terms of ψ as $\boldsymbol{B} = \hat{z} \times \boldsymbol{\nabla}\psi + B_z\hat{z}$.

Figure 2 presents sample numerical results for the island coalescence instability simulation with $\eta = 5 \cdot 10^{-5}$, $\eta_H = 2.4 \cdot 10^{-9}$, and $d_i = 2 \cdot 10^{-2}$. In particular, Figs. 2a, 2b, and 2c show profiles of the out-of-plane current density J_z at $t = 0$, and J_z and the out-of-plane magnetic field B_z at $t = 6.6(L/v_A)$ (the moment of maximum E_z), respectively. Attracting magnetic islands in Figs. 2a and 2b correspond to currents in the negative z direction. The positive, highly peaked current in Fig. 2b indicates the reconnection region. Physically, this current is induced to preserve magnetic flux in the $y - z$ plane [34]. The magnetic field B_z shown in Fig. 2c clearly possesses a quadrupolar structure, with red and blue colors indicating positive and negative values, respectively. Note that B_z is strongly peaked and practically constant along the island's magnetic separatrix.

We performed thirty five nonlinear simulations of the coalescence instability with parameters d_i, η, and η_H varying over wide ranges to describe situations corresponding to electron, Hall, and resistive MHD regimes: $0 \leq d_i \leq 1$, $0 \leq \eta \leq 10^{-3}$, and $0 \leq \eta_H \leq 7.6 \cdot 10^{-7}$. Situations when either resistive or hyper-resistive effects dominate are studied, as well as those where both are comparable. Values of δ, w, B_x^{in}, and E_z are obtained from the simulations at the time of maximum E_z. Values of δ and w are measured in two independent ways: by the half-width of the induced out-of-plane current J_z and by the maximum of the incoming and outgoing electron (or when appropriate ion) flow velocities. Both measurements are usually (but not always) possible and are normally rather consistent. B_x^{in} is measured at $(x, y) = (0, \delta/2)$. The numerical results are summarized in the Table, with the numbers in parentheses

obtained at the positions of the maximum flows and the numbers not in parentheses obtained at the half-width of J_z. To arrive at the results discussed next averages of the two numbers are used when appropriate.

Numerical results for the aspect ratio $\xi = \delta/w$ of the electron diffusion region versus d_*, and comparison with analytical predictions, are shown in Fig. 3. Results with $d_* \gtrsim 3$ correspond to the electron MHD regime, while those with $d_* \lesssim 1$ correspond to the resistive MHD regime. Here and elsewhere, results are coded according to the value of d_i, with the same symbols corresponding to the same d_i but to different values of η and/or η_H.

Figure 3a shows our "raw" simulation results. The largest value of ξ, $\xi \approx 0.53$, occurs for $d_i = 1$, while the smallest value of ξ, $\xi \approx 0.012$, occurs for $d_i = 3 \cdot 10^{-4}$. While we expect moderate values of ξ in the electron MHD regime [37, 44, 45], values of $\xi \sim 0.3 - 0.4$ (corresponding to open X-point geometries) are also found in the resistive MHD regime. Such ξ correspond to hyper-resistivity dominated cases. Recalling that $\xi \approx \sqrt[4]{16\eta_H/(B_x^{in} w^3)}$ for these cases, and using simulation parameters and data, $\eta_H \sim 10^{-7}$ and $B_x^{in} \sim w \sim 0.1$, gives $\xi \sim 0.36$, consistently with the simulation results.

Evaluating ξ from Eq. (23), which is trivial when hyper-resistivity is negligible but requires solving a cubic equation for ξ^2 in the general case, we obtain theoretical predictions ξ^{th}. Ratios of the numerical results over the theoretical predictions, ξ/ξ^{th}, are shown versus d_* in Fig. 3b, and demonstrate excellent agreement between the theory and the numerical experiments. The fact that the average value for ξ/ξ^{th} is slightly above unity can be attributed to our neglecting numerical factors of order unity in the discrete equations. The spread about the average value reflects errors in obtaining ξ from the simulations, and the effect of various approximations of the

analytical model.

Figure 4 shows numerical results for reconnection rates corresponding to values of ξ shown in Fig. 3, and compares them with analytical predictions from Eq. (25). Color-coding agrees with that used in Fig. 3. Figure 4a shows maximum reconnection rates E_z versus d_i as directly obtained from the numerical simulations. One can see that E_z is small and independent of d_i for $d_i \lesssim 10^{-2}$ (resistive MHD regime), but becomes much larger and proportional to d_i for $d_i \gtrsim 0.1$ (electron MHD regime). The scaling $E_z \propto d_i$ is shown with a solid black line. This agrees with our prediction $E_z = d_i(B_x^{in})^2/w$ provided that w and B_x^{in} are independent of d_i (the case for the island coalescence instability). We note that, depending on the macroscopic driver, situations might exist where w and B_x^{in} scale with d_i, and hence $E_z \propto d_i^\beta$, $\beta \neq 1$ [38, 39].

Figure 4b shows the same numerical data, but normalized, i.e., E_{z*} versus d_*. The normalized reconnection rates can vary over orders or magnitude for fixed values of d_i, depending on values of η and η_H, are generally small for $d_* \lesssim 1$ (resistive MHD regime), and are much larger for $d_* \gtrsim 3$ (electron MHD regime). The fact that E_{z*} is somewhat larger than obtained in particle simulations for electron MHD regimes is a consequence of our neglect of electron inertia. The ratio of the largest to the smallest reconnection rates in Fig. 4b is $1.2 \cdot 10^3$. Notice that relatively large $\xi \sim 0.3 - 0.4$, produced by hyper-resistivity in resistive MHD regime and corresponding to open X-point configurations, do not result in fast magnetic reconnection. At the same time, fast magnetic reconnection is possible for elongated electron diffusion regions, as follows from $\xi \propto S^{-1}$ in electron MHD (see also Ref. [21]), and as has been observed in recent particle simulations [36]. This suggests that the X-point geometry alone does not determine the reconnection rate.

Finally, Fig. 4c shows the ratio of E_{z*} to the analytical prediction, E_{z*}^{th}, from Eq. (25). We obtain $0.5 < E_{z*}/E_{z*}^{th} < 2.3$, i.e., we under- or over-predict the numerical reconnection rate by no more than a factor of two. This is still remarkable given the $1.2 \cdot 10^3$ spread in "raw" E_{z*} values.

9 INTUITIVE PICTURE FOR FAST RECONNECTION

Fast magnetic reconnection in Hall MHD can be easily understood from the following intuitive physical picture. Neglecting hyper-resistivity for simplicity, the reconnection rate $E_z = -\eta J_z|_{x=y=0} \sim \eta B_x^{in}/\delta$. Consequently, E_z is determined by δ.

In resistive MHD (Sweet-Parker) reconnection, electrons and ions do not separate and green and grey rectangles in Fig. 1 coincide. In particular, $B_x^{in} = B_x^{out}$ and $w = \Delta$. The diffusion region thickness δ can be easily estimated by balancing the sink and the source for B_x^{out} in Faraday's law, Eq. (1): $\eta B_x^{out}/\delta^2 \sim V_y^{out} B_x^{out}/\delta$. Using continuity of the ion flow, $V_y^{out}\Delta = V_x^{out}\delta$, and noticing that the reconnected magnetic field energy is transferred into the ion flow energy, $V_x^{out} \sim B_x^{out}$, gives $\delta \sim \delta_{SP} \equiv \sqrt{\Delta\eta/B_x^{out}} = [\sqrt{2}\Delta/S_\eta]^{1/2}$. This results in slow Sweet-Parker reconnection with $E_{z*} \sim [\eta/(B_x^{out}\Delta)]^{1/2} = [\Delta S_\eta/\sqrt{2}]^{-1/2}$.

Electrons and ions do separate in Hall MHD reconnection. Again, balancing the sink and the source for B_x^{in} in Faraday's law, we obtain $\eta B_x^{in}/\delta^2 \sim V_{e,y}^{in} B_x^{in}/\delta$. Employing continuity of the electron flow, $V_{e,y}^{in} w = V_{e,x}^{in}\delta = V_{e,x}^{out} d_i$, recalling that electron and ion flows coincide at the ion inertial region boundary, $V_{e,x}^{out} \approx V_x^{out}$, and noticing that the reconnected magnetic field energy is transferred into the ion flow energy, $V_x^{out} \sim B_x^{in}$, gives $\delta \sim \delta_H \equiv w\eta/(\sqrt{2}B_x^{in} d_i) = w/(d_i S_\eta)$. This results in

resistivity-independent, fast reconnection with $E_{z*} \sim d_i/w$. Transition between the two regimes clearly occurs at $\delta_{SP} \sim \delta_H$, or equivalently at $\delta_{SP} \sim d_i$. The preceding argument can be generalized for finite hyper-resistivity.

10 CONCLUSIONS

This work addresses a long-standing problem of basic plasma physics by presenting, for the first time, a simple analytical theory and an intuitive physical picture for magnetic reconnection in Hall MHD with arbitrary ion inertial length d_i without a guide field. The main results of the theory, namely the aspect ratio of the electron diffusion region and the reconnection rate, are given by Eqs. (23) and (25), respectively. Our theory recovers the well-known slow Sweet-Parker reconnection in the resistive MHD limit ($d_i \ll \delta$) and confirms long-standing numerical results by predicting fast reconnection and its independence of the dissipation mechanism in Hall ($d_i \sim \delta$) and electron ($d_i \gg \delta$) MHD. The theory also describes the transition between different regimes, which occurs when $d_i \sim \delta$, and is thoroughly benchmarked with two-dimensional numerical simulations.

Our theory resolves a number of existing controversies in the field of magnetic reconnection. In particular, it predicts that fast reconnection is possible for both open X-point and elongated diffusion region configurations, and that the reconnection rate can have a strong scaling with d_i, as is the case for the magnetic island coalescence instability. In addition, both the theory and the simple intuitive physical picture for fast reconnection do not employ fast dispersive waves. This fact provides a hint that a popular concept [44, 45] of fast dispersive waves being instrumental for fast reconnection may not be valid. Indeed, both numerical [46, 47, 48] and analytical

[24] studies of electron-positron plasmas, which do not support fast dispersive waves, show that such plasmas nevertheless exhibit fast magnetic reconnection.

Finally, the theory presented herein provides simple benchmarks for existing and future fluid codes, and allows quick estimates of the reconnection rate and of certain parameters of the microscopic reconnection region from experimental data. More importantly, our approach provides a flexible, extendable framework for the study of magnetic reconnection in a fluid context. For example, it has been be modified [24] to understand recent numerical claims of fast reconnection in electron-positron plasmas [46, 47, 48]. Many additional physical effects, such as electron inertia, ion viscosity, guiding magnetic field, and perhaps certain kinetic effects, can be considered, and their influence on magnetic reconnection assessed.

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FIGURE CAPTIONS

Figure 1. Reconnection region geometry.

Figure 2. Profiles of J_z and B_z from the numerical simulation of the island coalescence instability with $\eta = 5 \cdot 10^{-5}$, $\eta_H = 2.4 \cdot 10^{-9}$, and $d_i = 2 \cdot 10^{-2}$. (a) J_z at $t = 0$. Attracting magnetic islands correspond to currents in the negative z direction. (b) J_z at the moment of maximum reconnection rate. The reconnection region corresponds to the large peaked induced current in the positive z direction. (c) Quadrupolar structure of B_z at the moment of maximum reconnection rate.

Figure 3. Aspect ratio, $\xi = \delta/w$, of the electron diffusion region versus normalized d_i. Results are coded according to the value of d_i, with the same symbols corresponding to the same d_i but to different values of η and/or η_H. $\square$ (red) correspond to $d_i = 0$, $\blacksquare$ (red) to $d_i = 10^{-3}$, $\bullet$ (brown) to $d_i = 2 \cdot 10^{-3}$, $\circ$ (brown) to $d_i = 4 \cdot 10^{-3}$, $\bigstar$ (black) to $d_i = 7 \cdot 10^{-3}$, $\Diamond$ (green) to $d_i = 8 \cdot 10^{-3}$, $\blacklozenge$ (green) to $d_i = 10^{-2}$, $\blacktriangleleft$ (yellow) to $d_i = 2 \cdot 10^{-2}$, $\blacktriangle$ (blue) to $d_i = 3 \cdot 10^{-2}$, $\triangle$ (blue) to $d_i = 9 \cdot 10^{-2}$, $\blacktriangledown$ (blue) to $d_i = 3 \cdot 10^{-1}$, and $\triangledown$ (blue) to $d_i = 1$. (a) "Raw" numerical results obtained from full two-dimensional simulations. (b) Ratio of the numerical results to analytical predictions from equation (23).

Figure 4. Reconnection rate versus d_i for situations shown in Fig. 3. (a) Numerical reconnection rates E_z. Solid black line corresponds to $E_z \propto d_i$. (b) Normalized numerical reconnection rates E_{z*} versus normalized d_i. (c) Ratio of numerical to analytical [from Eq. (25)] reconnection rates.

TABLE. Values of discrete variables from numerical simulations.

d_i	η	η_H	δ	w	B_x^{in}	E_z
0	0	10^{-7}	0.033(0.038)	0.1(0.15)	0.36(0.38)	0.020
	10^{-5}	10^{-12}	0.0024	0.21	0.50	0.0053
	10^{-4}	10^{-12}	0.0068	0.16	0.39	0.014
0.001	0	10^{-8}	0.017(0.016)	(0.086)	0.42(0.41)	0.016
	0	10^{-7}	0.034(0.032)	0.11(0.078)	0.35(0.34)	0.019
0.002	0	10^{-8}	0.017(0.016)	(0.078)	0.42(0.41)	0.016
	0	10^{-7}	0.030(0.034)	(0.078)	0.35(0.33)	0.019
	10^{-5}	$2.4 \cdot 10^{-10}$	0.0066(0.0062)	(0.16)	0.49(0.47)	0.0079
	$2 \cdot 10^{-5}$	$2.4 \cdot 10^{-10}$	0.0069(0.0067)	(0.16)	0.49(0.48)	0.0091
	$5 \cdot 10^{-5}$	$2.4 \cdot 10^{-10}$	0.0075(0.0068)	(0.14)	0.45(0.42)	0.012
	10^{-4}	$2.4 \cdot 10^{-10}$	0.0085(0.0072)	0.13(0.13)	0.41(0.37)	0.015
	$2 \cdot 10^{-4}$	$2.4 \cdot 10^{-10}$	0.011(0.0095)	0.11(0.10)	0.35(0.334)	0.019
0.004	0	10^{-8}	0.016(0.016)	(0.078)	0.41(0.40)	0.017
	0	10^{-7}	0.034(0.030)	0.10(0.078)	0.35(0.33)	0.019
0.007	$2 \cdot 10^{-5}$	$8.5 \cdot 10^{-10}$	0.0080(0.0084)	(0.10)	0.41(0.42)	0.013
	$5 \cdot 10^{-5}$	$8.5 \cdot 10^{-10}$	0.0082(0.0084)	0.11(0.10)	0.40(0.40)	0.015
	10^{-4}	$8.5 \cdot 10^{-10}$	0.0087(0.0084)	0.10(0.094)	0.36(0.35)	0.018
	$2 \cdot 10^{-4}$	$8.5 \cdot 10^{-10}$	0.010(0.0094)	0.091(0.081)	0.32(0.30)	0.021
	$5 \cdot 10^{-4}$	$8.5 \cdot 10^{-10}$	0.017(0.013)	0.088(0.073)	0.26(0.22)	0.023
0.008	0	10^{-7}	0.033(0.030)	(0.07)	0.34(0.32)	0.020

TABLE (continued). Values of discrete variables from numerical simulations.

d_i	η	η_H	δ	w	B_x^{in}	E_z
0.01	$2 \cdot 10^{-5}$	$1.2 \cdot 10^{-9}$	0.0077(0.0084)	0.088(0.079)	0.37(0.38)	0.017
	$5 \cdot 10^{-5}$	$1.2 \cdot 10^{-9}$	0.0079(0.0084)	0.087(0.079)	0.34(0.36)	0.019
	10^{-4}	$1.2 \cdot 10^{-9}$	0.0084(0.0084)	0.083(0.078)	0.32(0.32)	0.020
	$2 \cdot 10^{-4}$	$1.2 \cdot 10^{-9}$	0.0096(0.0094)	0.078(0.069)	0.29(0.29)	0.022
	$5 \cdot 10^{-4}$	$1.2 \cdot 10^{-9}$	0.016(0.013)	0.095(0.066)	0.25(0.22)	0.023
	10^{-3}	$1.2 \cdot 10^{-9}$	0.030(0.021)	0.12(0.070)	0.21(0.17)	0.023
0.02	10^{-4}	$2.4 \cdot 10^{-9}$	0.0080(0.0073)	0.050(0.042)	0.23(0.22)	0.026
	$2 \cdot 10^{-4}$	$2.4 \cdot 10^{-9}$	0.0092(0.0084)	0.052(0.045)	0.22(0.21)	0.026
	$5 \cdot 10^{-4}$	$2.4 \cdot 10^{-9}$	0.014(0.012)	0.062(0.052)	0.20(0.19)	0.025
0.03	0	10^{-7}	0.027(0.026)	0.082(0.062)	0.26(0.26)	0.027
0.09	0	10^{-7}	0.018(0.023)	0.049(0.047)	0.14(0.15)	0.049
0.3	0	10^{-7}	0.010(0.016)	0.031(0.031)	0.11(0.12)	0.18
1	10^{-5}	$7.6 \cdot 10^{-7}$	0.014(0.018)	0.042(0.040)	0.15(0.16)	0.35
	10^{-5}	$2.3 \cdot 10^{-6}$	0.024(0.028)	0.058(0.062)	0.17(0.19)	0.39
	10^{-4}	$7.6 \cdot 10^{-6}$	0.040(0.063)	0.11(0.086)	0.15(0.15)	0.57

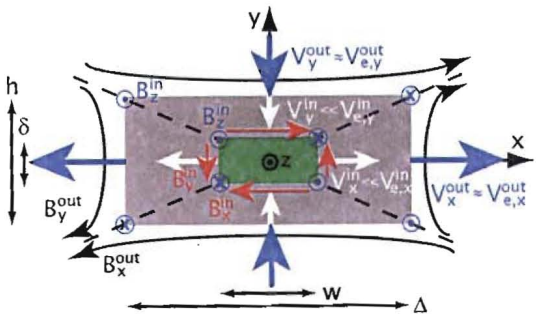


Figure 1.

Simakov and Chacon
Quantitative analytical model...

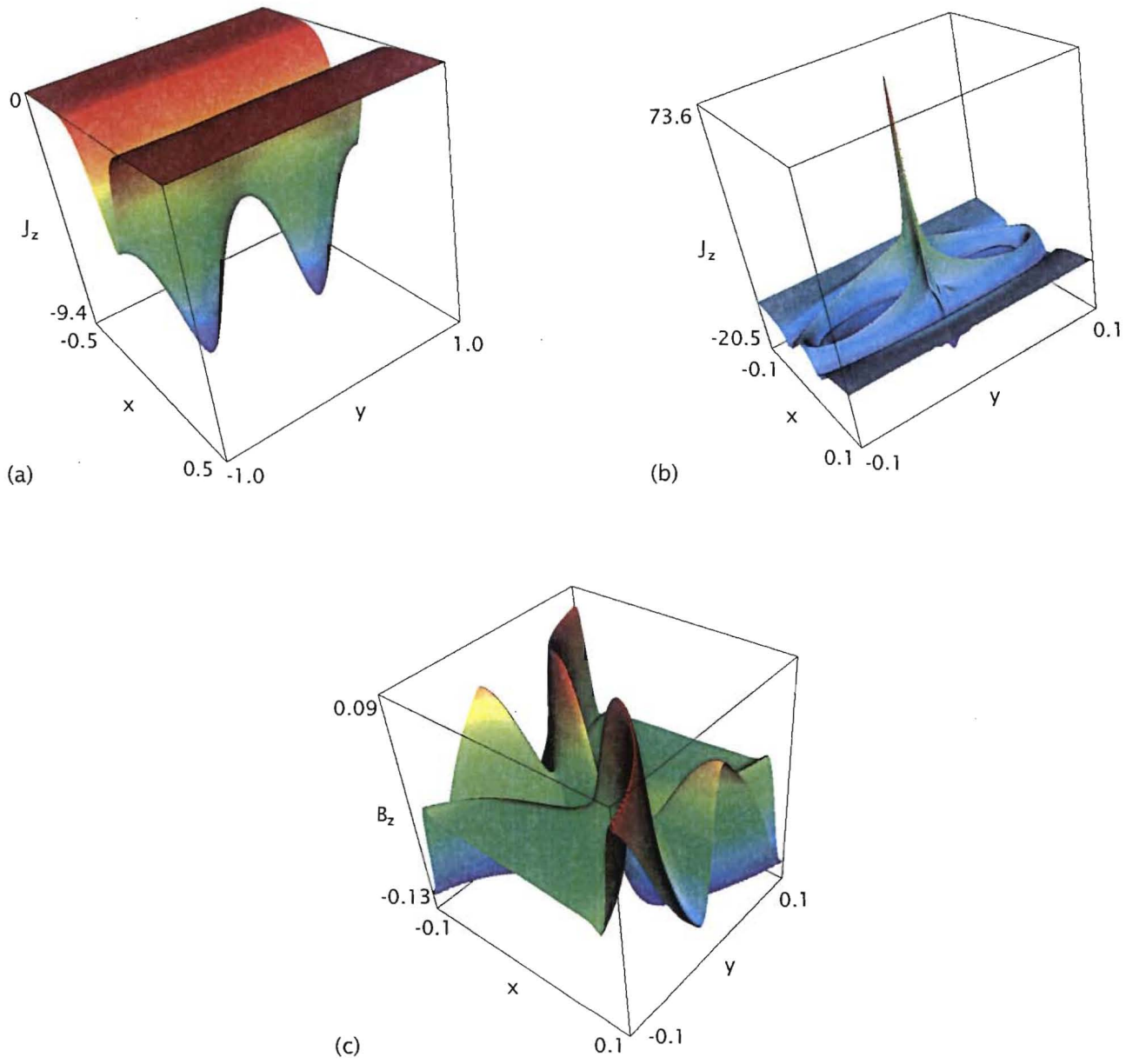


Figure 2.

Simakov and Chacon
Quantitative analytical model...

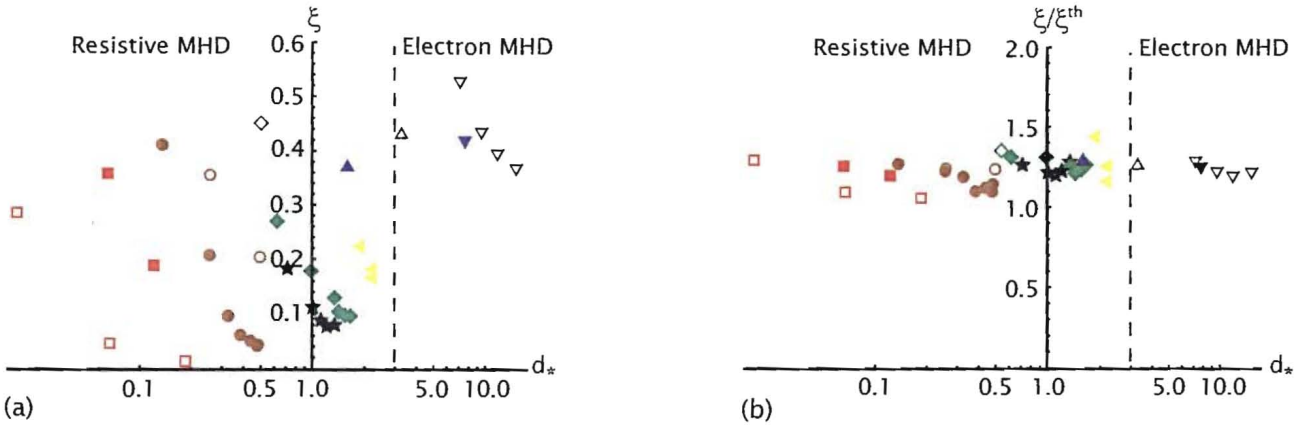


Figure 3.

Simakov and Chacon
Quantitative analytical model...

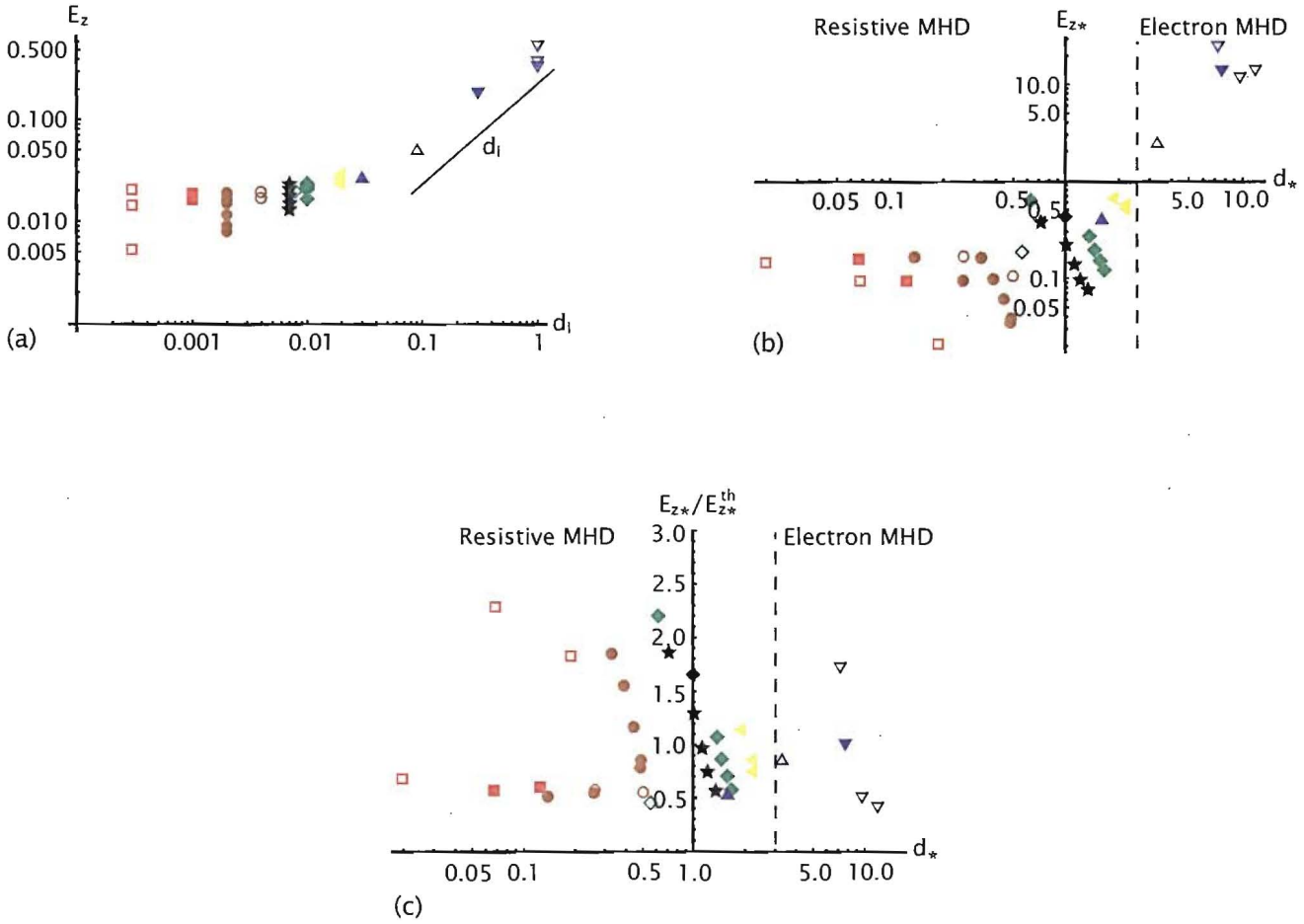


Figure 4.

Simakov and Chacon
Quantitative analytical model...